**KENYA MEDICAL TRAINING COLLEGE**
**FACULTY OF REHABILITATIVE SCIENCES**
**DEPARTMENT OF MEDICAL ENGINEERING**

**FINAL QUALIFYING EXAMINATION**
**FOR**
**CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY**

**PAPER: ELECTRICAL ENGINEERING THEORY**

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|---|---|
| **DATE: 23rd July 2021** | **TIME: 3 HOURS (9:00AM – 12:00 NOON)** |

**INSTRUCTIONS**

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

## SECTION 1

1. Which one of the following is not a basic element of a transformer?
   A. Primary winding
   B. Secondary winding
   C. Core
   D. Mutual flux
2. Core is used in a transformer so as to
   A. Prevent eddy currents
   B. Decrease reluctance of common magnetic circuit
   C. Eliminate magnetic hysteresis
   D. Decrease iron losses
3. Lamination of transformers parts helps to achieve
   A. Reduce hysteresis loss
   B. Minimize eddy currents loss
   C. Reduce cost
   D. Simplify its construction
4. A transformer has 1000 turns on the primary side and its connected to a 250V AC supply. For a secondary voltage of 400V to be realized, what should be the number of secondary turns?
   A. 250
   B. 400
   C. 1600
   D. 1250
5. In a transformer, the leakage flux of each winding is proportional to current in that winding because
   A. Ohm's law applies to magnetic circuits
   B. Leakage paths do saturate
   C. The two windings are electrically isolated
   D. Leakage paths do not saturate
6. When performing short circuit test in a transformer
   A. High voltage side is usually short circuited
   B. Low voltage side is usually short circuited
   C. Any side is short circuited with preference
   D. Both low and high voltage sides are short circuited
7. The main purpose of performing open-circuit test on a transformer is to measure its
   A. Core loss
   B. Copper loss
   C. Total loss
   D. Insulation resistance
8. An autotransformer has transformation ratio of 0.8 and supplies a 3KW load. What is the power transferred from the primary to the secondary?
   A. 0.6 KW
   B. 1.5 KW
   C. 0.27 KW
   D. 2.4 KW

9. Three phase alternators are usually connected in star because
   A. Higher terminal voltage is obtained
   B. Magnetic losses are minimized
   C. Smaller conductors can be used
   D. Less turns of wires are required
10. Armature reaction in an alternator primarily affects
   A. Rotor speed
   B. Terminal voltage per phase
   C. Generated voltage per phase
   D. Frequency of armature current
11. What is the role of damper winding in a synchronous motor?
   A. Develop necessary starting torque
   B. Control the power factor of the motor
   C. Stabilize rotor motion and suppress oscillations
   D. Limit motor current.
12. An electric motor in which both the rotor and the stator field rotate at the same speed is called _____ motor.
   A. Synchronous
   B. DC
   C. Universal
   D. Induction
13. The direction of rotation of a synchronous motor can be reversed if
   A. Supply phase sequence is reversed
   B. Current to field winding is reversed
   C. Polarity of rotor poles is reversed
   D. All of the above
14. The field of a given synchronous motor is under excited. The power factor will be
   A. Leading
   B. Lagging
   C. Unity
   D. More than unity
15. In a synchronous motor, the copper losses are met by
   A. Motor input
   B. Armature input
   C. Supply lines
   D. DC source
16. The principle of operation of three phase induction motor is most similar to that of
   A. Synchronous motor
   B. Transformer with a shorted secondary
   C. Repulsion start induction motor
   D. Capacitor start induction motor
17. Magnetizing current drawn by induction motors results to _____ power factor.
   A. Unity
   B. Leading
   C. Lagging
   D. Zero

18. Increasing air gap in an induction motor will lead to
    A. Increase in power factor
    B. Reduction in speed
    C. Increase in air gap flux
    D. Increase in magnetizing current
19. The slip speed in an induction motor may be defined as the
    A. number of pairs of poles ÷ frequency
    B. rotor speed − synchronous speed
    C. rotor speed + synchronous speed
    D. synchronous speed − rotor speed
20. What replacement is required to convert an AC generator to DC generator?
    A. Concave magnets with horseshoe magnets
    B. Armature with coil
    C. Slip rings with split rings
    D. All of the above
21. Which of the following is not a component of AC generator?
    A. Field
    B. Armature
    C. Prime mover
    D. Starter
22. In an AC generator, the value of voltage generated depends on
    A. The number of turns in the coil
    B. The number of coils
    C. Strength of the field
    D. All of the above
23. The output voltage of a three phase alternator is regulated by the
    A. AC voltage to armature
    B. AC voltage to field
    C. DC voltage to armature
    D. DC voltage to field
24. AC and DC generators are similar in that they
    A. Both generate alternating voltages
    B. Both rectify the voltage before delivery
    C. Are constructed of same physical size for the same kilowatt rating
    D. Both supply
25. The field winding in a DC machine is mounted on
    A. Rotor
    B. Stator
    C. Prime mover
    D. Brushes
26. In spite of heavy initial investments, DC motors are used due to
    A. Lower losses
    B. Improved power factor of the system
    C. Flexible and ease of control
    D. All of the above

27. Which one of the following does not influence the capacitance of a capacitor?
    A. Plate area
    B. Plate separation
    C. Nature of the dielectric
    D. Plate thickness
28. A current is said to be alternating when it changes in
    A. Magnitude only
    B. Direction only
    C. Both magnitude and direction
    D. Phases
29. In an AC circuit, a low value of KVAR compared to KW indicates
    A. Low efficiency
    B. Unity power factor
    C. High power factor
    D. Maximum load current
30. Root mean square value of current is equal to
    A. 1.414 times the maximum value
    B. 0.707 times the maximum value
    C. 1.732 times the maximum value
    D. Twice the maximum value
31. Which one of the following is not a method of neutral grounding?
    A. Resistance grounding
    B. Equipment grounding
    C. Solid grounding
    D. Reactance grounding
32. What is the appropriate current rating of a cooker unit circuit breaker?
    A. 15A
    B. 25A
    C. 32A
    D. 63A
33. Earth electrode provides
    A. High resistance
    B. Medium resistance
    C. Low resistance
    D. Very high resistance
34. Strength of an electromagnet can be increased by
    A. Increasing the cross-sectional area
    B. Increasing the number of turns
    C. Increasing current supply
    D. All of the above
35. In the Fleming's left hand rule, fore finger always represents
    A. Voltage
    B. Current
    C. Magnetic field
    D. Direction of force on a conductor

36. The purpose of a commutator in a DC generator is to
    A. Increase output voltage
    B. Reduce sparking at brushes
    C. Provide smoother output
    D. Convert the induced AC in to DC
37. The critical resistance of a DC generator is the resistance of
    A. Armature
    B. Filed
    C. Load
    D. Brushes
38. In a split phase motor, the running winding should have
    A. High resistance and low inductance
    B. Low resistance and High inductance
    C. High resistance and high inductance
    D. Low resistance and low inductance
39. In a capacitor start and run motors, the function of the run capacitor in series with the auxiliary winding is to
    A. Improve power factor
    B. Increase overload capacity
    C. Reduce fluctuations in torque
    D. To improve torque
40. In a split phase motor
    A. The starting winding is connected through a centrifugal switch
    B. The running winding is connected through a centrifugal switch
    C. Both start and run windings are connected through a centrifugal switch
    D. Centrifugal switch is used to control supply voltage

## SECTION 2

1. a) Differentiate between separately excited and self-excited DC generators. (4 marks)
   b) Which type of single phase AC motor always has relatively high power factor. (1 mark)

2. a) State **three** types of DC motors (3 marks)
   b) Define *armature reaction.* (2 marks)

3. A DC shunt generator is used to supply a current of $20A$ at $30V$ to a load. The armature resistance is $0.7\Omega$ and the field resistance is $60\Omega$. Assuming brush drop of $2V$, calculate:
   i. Armature current (3 marks)
   ii. Generated voltage (2 marks)

4. a) Define *commutation.* (2 marks)
   b) State the **two** effects of armature reaction. (2 marks)
   c) A universal motor can run on either _____ or _____ supply. (1 mark)

5. a) Define the term power factor as used in AC circuits. (2 marks)
   b) State **three** reasons why electricity consumers are encouraged to operate at high power factor. (3 marks)

6. Explain why single phase induction motor is not self-starting. (5 marks)

7. How do you modify single phase induction motor to make it self-starting. (5 marks)

8. a) State **any two** advantages of three phase power system (2 marks)
   b) Define the following terms and give their respective units:
      i. Impedance
      ii. Admittance (3 marks)

## SECTION 3

1. a) State **any two** advantages and disadvantages of three phase induction motors. (4 marks)
   b) State **any three** properties of an ideal transformer. (3 marks)
   c) Explain how a slip ring motor is started. (6 marks)
   d) Draw a well labelled diagram of a two pole DC machine. (7 marks)